

A Beginner Guide to Debugging Slow Web Applications

Debugging slow web applications can feel overwhelming when you do not know where to look first. Learning a few basic commands helps you isolate performance bottlenecks in seconds instead of guessing.

What you'll learn:

- 1 Measuring Server Response Time with cURL
- 2 Checking Server Resource Spikes with htop
- 3 Streaming Real-Time Logs with tail
- 4 Inspecting Active System Connections with ss

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1

TIP 1 OF 4

Measuring Server Response Time with cURL

When an application feels slow, the first step is knowing whether the delay comes from the network or the server. You can use cURL to format timing output and get exact statistics on DNS lookup, connection time, and total transfer duration.

```
$curl -w "DNS: %{time_namelookup}s | Connect: %{time_connect}s | Total: %{time_total}s\n" -o /dev/null
```

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2

TIP 2 OF 4

Checking Server Resource Spikes with htop

A web app often slows down because the underlying host is running out of memory or CPU cycles. Running htop displays an interactive, colorful view of CPU cores, RAM usage, swap space, and running processes sorted by resource consumption.

```
$sudo apt update && sudo apt install htop -y  
htop
```

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3

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Streaming Real-Time Logs with tail

Instead of opening huge static log files, stream updates directly to your terminal screen as users click around your site. The tail command with the follow flag lets you observe error codes, warning messages, and slow response paths live.

```
$tail -f -n 50 /var/log/nginx/error.log
```

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4

TIP 4 OF 4

Inspecting Active System Connections with ss

If your web app refuses to load or drops requests, your server might have exhausted its connection pool limits. The socket statistics command helps you inspect established connections and listening ports to confirm network health.

```
ss -s
```


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Key Takeaways

- + Always reproduce the issue using curl or browser DevTools first.
- + Check CPU and RAM utilization before tweaking application code.
- + Filter log output using grep to find relevant error status codes quickly.
- + Test endpoint speeds from multiple networks to isolate network issues.
- + Document baseline performance metrics so you know what normal looks like.

Watch Out For

- ! Restarting the server immediately, which wipes out process logs and live evidence.
- ! Assuming the backend is slow when uncompressed frontend assets are actually the problem.
- ! Changing multiple configuration settings at once instead of testing one fix at a time.

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